

MESSENGINFO

Ukrainian Document Translation Engine v5.0

Final Controlled Autonomy Engineering Standard

Purpose: Build an AI-assisted, human-certified translation pipeline for Ukrainian documents → English for USCIS / official use.

Audience: Product, engineering, QA, legal/compliance, UX, and translation reviewers.

Status: Final consolidated standard for pilot implementation.

Core rule: A translation is not finished when it looks good. It is finished only when every critical field is traceable back to the original document and the final certification is signed by a competent human.

Item	Decision
Operating model	Controlled autonomy: AI drafts, validates, and renders; humans confirm critical fields and sign.
Primary risk to avoid	Beautiful but wrong translation: incorrect number, date, name, agency, address, or over-certified scope.
Pilot principle	Keep MVP narrow, auditable, and shippable: passport booklet + basic civil documents first.
Provider policy	Provider-agnostic architecture. DeepSeek-first for AI/API testing unless explicitly changed. Vision provider remains pluggable.

1. Executive Decision

This v5.0 standard consolidates the previous Translation Brain, Numeric Accuracy Protocol, v3 Controlled Autonomy architecture, and v4 business/UX protections into one implementable engineering specification. It intentionally avoids the false target of “100% autonomous certified translation.” The correct target is controlled autonomy: automation handles the repetitive work, while critical facts and certification remain human-accountable.

Decision	Standard
Do	Automate OCR/vision, extraction, normalization, evidence crops, QA validators, source-to-final audit, PDF/DOCX rendering, and payment gating.
Do not	Let AI certify, guess low-confidence fields, modernize historical terms, promise USCIS acceptance, or produce final PDF without source trace.
MVP boundary	Start with Ukrainian passport booklet, ID card, birth certificate, marriage certificate, and simple one-page certificates. Defer military, transcripts, and complex tables unless manual review is available.
Business boundary	Protect final A4/PDF output behind payment, but allow enough field-level preview and evidence review before payment so users trust the service.

2. Product Philosophy and Legal Posture

- The service is a document-preparation and translation-draft tool, not a law firm and not immigration legal advice.
- The system may prepare an AI-assisted draft. A translation becomes certified only after a competent human reviews and signs the certification of translation accuracy.
- All public copy must separate: AI draft, user-reviewed translation, human translator reviewed, certified translation, and notarized signature.
- The service must not claim “USCIS accepted,” “guaranteed,” “approved translation,” “certified by AI,” or “instant certified translation.”
- The safe public wording is: “Prepared for USCIS-style submission,” “AI-assisted draft for human review,” and “You review and sign if competent.”

Status	Meaning
AI Draft	Machine-assisted draft. Not certified. Must be reviewed.
User-reviewed	User confirmed field values. Not automatically a certified translation unless user signs the certification and is competent.
Human translator reviewed	A competent human translator checked/corrected the output.
Certified	A named human signs a certification of accuracy and competence.
Notarized	Notary verifies identity/signature, not translation quality.

3. User Model: 35-80-Year-Old Non-Lawyer

The default user is not a lawyer, may be anxious, may not understand USCIS language, and may struggle with mobile uploads, glare, cropping, typing, and legal distinctions. The system must behave like a patient translation office clerk: direct, large, simple, and forgiving.

Area	Requirement
Screen design	One screen = one task. Avoid dense tables in the user-facing flow. Use 18px+ body text, 24px+ headers, strong contrast, and 44px+ tap targets.
Photo help	Specific plain-language instructions: “The lamp light covers the date in the lower left. Please retake near a window.” Not “OCR failed.”
Evidence review	Show one extracted field beside the cropped source image. Buttons: “Correct” / “Fix”.
Fallback	If user is stuck, provide Telegram/support/manual review path. Do not trap the user in endless retakes.
Trust	Explain what happens before payment, after payment, and who signs the final certification.

4. Controlled Autonomy Architecture

Stage	Module	Function
1. Intake	Smart Retake Engine	Quality gate, document type estimate, glare/crop/blur detection, anti-loop retake flow.
2. Session memory	Packet State Manager	Maintains identity anchors, filing spelling, uploaded docs, scope, corrections, and payment state.
3. Extraction	Vision/OCR + Numeric Extractor	Extracts raw fields with bbox/source zone; does not normalize prematurely.
4. Knowledge	RAG Glossary Loader	Loads only relevant glossary modules: passport, civil status, military, medical, education, historical geography.
5. Linguistic	Linguistic Normalizer	Duplicate collapse, nominative case restoration, transliteration, agency resolution, admin terms.
6. QA	Validator Suite	Numeric, dates, source trace, forbidden phrases, scope, identity consistency, service claims.
7. User review	Evidence Viewer	User confirms or corrects critical fields from visible crop cards.
8. Billing	Payment Gate	Unlocks final A4/PDF/DOCX; pre-payment only field cards/watermarked preview.
9. Certification	Certification Record	Collects signer identity, competence statement, typed signature, date, optional contact details.
10. Rendering	Bureau-Style Renderer	Generates clean translation + certification + original pages; no internal QA notes.

5. End-to-End Pipeline

1. Create TranslationSession and Packet State.
2. User selects task/document type or chooses “I am not sure.”

3. Collect identity anchor if available: international passport, I-94, USCIS notice, EAD, or manually entered controlling spelling. If none exists, use official Ukrainian transliteration after review.
4. Upload source document photo/PDF. Run image-quality gate before paid rendering.
5. Classify document and create zone map. Never translate immediately.
6. Extract raw fields with source_label, source_zone, bbox, language_layer, confidence, and raw text.
7. Run numeric double-pass for dates, document identifiers, record numbers, addresses, and case/account numbers.
8. Normalize using controlled glossary, transliteration rules, historical geography lock, and nominative case restorer.
9. Run validators and build QA Evidence Table.
10. Show Evidence Viewer to user for critical fields. Capture corrections through Correction Classifier.
11. Gate final rendering by: source trace, critical field confidence, user/human confirmation, certification record, and payment success.
12. Render final bureau-style PDF/DOCX with original pages attached separately.
13. Write audit log and version/redline record. Add safe confirmed corrections to Translation Memory only after classification.

6. Translation Brain: Content, Not Layout

Rule	Meaning
Translate	Document title, series, number, names, dates, places, issuing authority, readable stamp text, registration data, marital-status entries, record numbers, and legally meaningful notes.
Do not translate	Cover, page visual labels, uploaded-image placeholders, “round seal,” emblems, ornaments, photo descriptions, bilingual layout, unreadable decorative marks, or scanner artifacts.
Exception	If the user explicitly requests a forensic/literal translation, a different template may include layout notes, but this is not the default USCIS/bureau-style output.

7. Source Hierarchy

14. Original document text physically tied to the correct source label and zone.
15. Readable official stamps, endorsements, and registration marks.
16. Controlling Latin spelling from official IDs or U.S. records.
17. Official Ukrainian transliteration when no controlling Latin spelling exists.
18. Controlled glossary / agency registry / historical geography registry.
19. Human review for unresolved abbreviations, low-confidence handwriting, cropped fields, or conflicting sources.

Earlier drafts, model memory, prior answers, and “what seems likely” are never source of truth.

8. Document Classification and MVP Boundary

Document type	Critical fields	Pilot status
Ukrainian internal passport booklet	Series, number, name, DOB, POB, sex, issuing authority, date of issue, registration, marital status, children, military duty if present.	MVP yes
Ukrainian ID card	Document number, name, DOB/POB, sex,	MVP yes

	issue/expiry dates, authority, record number.	
International passport	Controlling Latin spelling, passport number, dates, authority, MRZ.	Anchor / MVP partial
Birth / marriage / divorce / death certificate	Act record number/date, parties, event date/place, issuing office.	MVP yes for simple layouts
Military document	Rank/status, military office, restrictions, record numbers.	Manual review first
Diploma / transcript	Institution, degree, qualification, tables, dates, grades.	Defer or manual review
Medical / vaccination record	Vaccines, dates, providers, lot numbers if present.	Defer unless glossary exists

9. Field-Based Extraction Model

Every extracted item must become a traceable field object. No source trace means no final output.

```
{
  "field": "Date of Issue",
  "document_type": "passport_booklet_ukraine_internal",
  "source_label": "Дата видачі паспорта",
  "source_zone": "issuance_block.date_line",
  "bbox": [x1, y1, x2, y2],
  "raw_value": "19 лютого 2003",
  "normalized_value": "19 February 2003",
  "language_layer": "uk",
  "confidence": 0.94,
  "passes": ["visual_pass_1", "visual_pass_2", "ocr_compare"],
  "review_required": false
}
```

10. Numeric Accuracy Protocol

Numbers are evidence. They are never inferred, guessed, “corrected by logic,” copied from memory, or borrowed from another field.

Numeric category	Required verification
Passport series/number	Separate perforation or printed identifier from normal OCR. Compare ambiguous digits. Expected format must match document type.
Dates	Tie date to printed label; read raw day/month/year; validate month dictionary; back-map English date to source.
Record / act numbers	Do not confuse with page number, registry number, certificate number, or passport number.
Addresses	Separate building, block, apartment, district, city/settlement, oblast. Do not Americanize.
Case/account numbers	Preserve exactly. Never insert spaces/hyphens unless source or official format requires them.

11. Zone-Lock and Date Handling

Field	Allowed source zone	Fail condition
Date of Birth	Line under Дата народження / Дата рождження	Date taken from issue block or stamp.
Date of Issue	Line under Дата видачі паспорта / Дата	Date taken from neighboring handwriting,

	выдачи паспорта	stamp, or previous draft.
Registration Date	Residence-registration stamp line	Date borrowed from issue page.
Photograph Attestation Date	Photo attestation/vkleyka-photo block only	Date confused with original issue date.
Passport Series/Number	Perforation or official identifier field only	Number parsed from background, page count, or prior answer.

UA	EN	UA	EN
січня	January	липня	July
лютого	February	серпня	August
березня	March	вересня	September
квітня	April	жовтня	October
травня	May	листопада	November
червня	June	грудня	December

12. Passport Booklet Series and Number Protocol

- Detect the full perforated sequence at the top/official identifier area.
- Split into series and number according to expected structure: commonly 2 letters + 6 digits for legacy internal passport booklets.
- Read each digit independently; compare ambiguous digit shapes inside the same sequence.
- Preserve the exact sequence; do not add punctuation, spaces, or corrections not present in the source.
- If any digit is uncertain after comparison, set review_required=true and block final rendering.

Digit risk	Comparator rule
8 vs 0	8 has a two-loop/double-pattern appearance; 0 has a single oval pattern. Compare against other 8s if present.
1 vs 7	Check vertical stroke and top mark; do not infer from number pattern.
6 vs 8/9	Check open/closed loop direction and tail.
Perforation + glare	Crop/zoom/deskew and perform second visual read.

13. Linguistic Intelligence Modules

Module	Requirement
Duplicate Language Collapser	Ukrainian/Russian duplicate labels become one English field. Ukrainian is primary; Russian is OCR support.
Nominative Case Restorer	Names in dative/genitive/instrumental must be restored to nominative before transliteration: Петренку Івану → Petrenko Ivan, not Petrenku Ivanu.
Controlling Spelling Resolver	Uses international passport/I-94/USCIS/EAD spelling when present; otherwise official Ukrainian transliteration.
Historical Geography Lock	Do not modernize historical names. If source says Kirovohrad, output Kirovohrad, not Kropyvnytskyi, unless document itself uses modern name.
Spelled-Out Numbers Parser	Parses dates and amounts written in words; stores raw phrase and normalized value in QA table.
Apostrophe/Soft Sign Rule	Ukrainian apostrophe and soft sign generally are not rendered as English punctuation under official transliteration: Дар'я → Daria.
Tabular Data Renderer	For transcripts/diplomas/medical tables, preserve tabular structure. Do not flatten into prose.

14. Names, Transliteration, and Identity Anchors

Priority	Source for Latin spelling
1	International Ukrainian passport / travel passport
2	USCIS notice, receipt, online account, or official U.S. immigration record
3	I-94
4	EAD / green card / state ID if relevant to packet
5	Official Ukrainian transliteration when no controlling Latin spelling exists

If the translation spelling and filing spelling differ, store both `translated_name` and `filing_name`, with `source_of_filing_name`. Do not silently switch variants across the packet.

15. Administrative Terms, Agencies, and Abbreviations

Source	Approved English
область	Oblast
район / р-н	Raion / District
місто / м.	city
смт / селище міського типу	urban-type settlement
село / с.	village
вул.	street
буд.	building / house no.
корп.	building block / block
кв.	apartment
ім.	named after
б/н	without number / No. not assigned
в/ч	Military Unit

Source abbreviation	Approved structured translation
МВС України	Ministry of Internal Affairs of Ukraine
ГУМВС України	Main Directorate of the Ministry of Internal Affairs of Ukraine
УМВС України	Directorate / Department of the Ministry of Internal Affairs of Ukraine
РВ УМВС	District Department of the Directorate of the Ministry of Internal Affairs
МВ УМВС	City Department of the Directorate of the Ministry of Internal Affairs
ДМС України	State Migration Service of Ukraine
РАЦС / ДРАЦС	Civil Registry Office / State Civil Registry Office

Forbidden unless explicitly present in a modern document: Police Department, passport police, local police office, Ukrainian police office.

16. Stamps, Seals, Signatures, and Empty Fields

Source condition	Output rule
Decorative seal/emblem, no readable text	Omit.
Readable seal or stamp text	Translate readable text only.
Visible but unreadable text	Use [illegible] or [partially illegible] only when legally necessary.
Signature image only	Usually omit in bureau-style output; do not write [signature]

	unless literal/forensic translation is requested.
Legally meaningful section blank and included	Use [blank] or [no entries].
Section not provided in uploaded pages	Do not invent it and do not imply the whole document was reviewed.

17. Scope Label and Bureau-Style Rendering

Source set	Safe title
All passport pages included and reviewed	English Translation of Passport of the Citizen of Ukraine
Only visible passport spread/page included	English Translation of the Provided Passport Pages / Extracted Passport Data
Single certificate page included	English Translation of [Document Type]

CERTIFIED ENGLISH TRANSLATION

[DOCUMENT TITLE]

Document Type:
[...]

Series / Number:
[...]

Surname:
[...]

Given Name:
[...]

Date of Birth:
[...]

Issued by:
[...]

Date of Issue:
[...]

CERTIFICATION OF TRANSLATION ACCURACY

I certify that I am competent to translate from Ukrainian/Russian into English and that this translation is complete and accurate to the best of my knowledge and ability.

Translator:
[Name]

Signature:

Rule	Values
Forbidden final output elements	Cover; Page 1/Page 2 as body text; Uploaded image; Round seal; Bilingual Ukrainian/Russian; Photo present; Original image; Translator Note; AI draft note; Looks like/probably/maybe/I think; Passport Holder's Signature: [signature] unless explicitly required.

18. Certification Record and Self-Certification UX

A checkbox alone is not a certification. It is only consent. For a USCIS-style package, the system must collect a certification record and render it in the final package according to the selected policy.

Item	Requirement
Required	Translator full legal name; language pair confirmation; certification statement; typed signature; signature date.
Policy-dependent	Address, phone, email may be shown visibly or stored in metadata/admin record depending on final compliance policy.
User flow	Simple language: “Sign only if you can read both languages and confirm the translation is complete and accurate.”
Gate	No certification record → no certified package.

19. Evidence Viewer and Human Critical Field Gate

Component	Requirement
Critical fields	Name, DOB, place of birth, document number, issue date, issuing authority, registration address, act/record number, certificate number.
Evidence card	Show field value + cropped source image + plain-language confidence note + buttons Correct/Fix.
Gate threshold	Critical field confidence below 0.95 requires user or human reviewer confirmation before final render.
User correction	Captured through Correction Classifier. Not all corrections become global memory.

20. Smart Retake Engine and Anti-Loop Policy

State	Action
Attempt 0	Accept if quality passes.
Attempt 1	Show specific retake instruction: glare/blur/crop/location.
Attempt 2	Show improved instruction + example image + option to upload a different photo.
Attempt 3	Offer manual review, continue with low-confidence warning, or support/Telegram fallback.

Do not create infinite retake loops. For older users, every failure must explain what went wrong in human language and provide an alternative path.

21. Payment Gate and Screenshot Protection

Gate	Rule
Before payment	Show field cards, evidence crops, correction UI, and possibly a blurred/watermarked layout preview. Do not show full clean A4/PDF/DOCX.
After payment	Unlock final bureau-style rendering and download.
Security	Watermark draft previews; blur critical data in full-layout previews; log payment state.

Trust balance	Do not hide everything before payment. User must be able to verify that the extracted data is correct.
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22. Translation Memory and Correction Classifier

Translation Memory is useful only if corrections are classified. Blindly learning from every user edit will spread errors.

Correction type	Memory behavior
controlling_spelling	Use for this person/packet; may become identity anchor.
OCR_error	Can become regression test and model improvement example.
glossary_update	Requires review before global glossary change.
one_document_exception	Applies only to this source document.
user_preference	Do not apply globally; may conflict with official spelling.
unsafe_correction	Block or require human review.

23. Source-to-Final Completeness Audit

- Compare original zones extracted → draft fields → user-confirmed fields → final rendered document → attached original pages.
- Fail if a source field exists but is missing in translation.
- Fail if final translation contains a field not found in source or human-approved packet context.
- Fail if number/date changed between review and final render.
- Fail if scope title is broader than uploaded pages.
- Fail if original pages are not attached to the final package.

24. Packet State Manager

```
{
  "session_id": "uuid",
  "user_language": "ru|uk|en",
  "global_identity": {
    "filing_name": "as provided by controlling source",
    "source_of_filing_name": "international_passport|I94|USCIS|EAD|manual|none",
    "date_of_birth": "YYYY-MM-DD|null",
    "review_status": "draft|user_confirmed|translator_confirmed"
  },
  "documents": ["uploaded_document_ids"],
  "field_decisions": ["confirmed critical fields"],
  "corrections": ["redline entries"],
  "payment_state": "not_started|pending|paid|failed|refunded",
  "certification_state": "missing|draft|signed",
  "render_state": "blocked|ready|rendered|downloaded"
}
```

25. Minimal API / JSON Contracts

Object	Minimum fields
TranslationSession	session_id, user_id, locale, status, payment_state, certification_state, render_state

UploadedDocument	document_id, session_id, file_path, type_guess, quality_status, pages, retention_until
ExtractedField	field_id, document_id, label, zone, bbox, raw_value, normalized_value, confidence, review_required
NumericEvidence	field_id, raw_value, normalized_value, pass_count, comparator_notes, confidence
IdentityAnchor	name_value, source_type, source_document_id, confirmation_status
UserCorrection	field_id, old_value, new_value, correction_type, reason, confirmed_by
CertificationRecord	name, language_pair, statement_version, signature, signed_at, optional_contact_fields
FinalRenderJob	session_id, template_version, input_field_hash, output_path, render_status
AuditLog	actor, action, resource, metadata, timestamp

26. Required Validators Before Final Output

Validator	Failure prevented
forbiddenPhraseValidator	Layout garbage and AI-looking output.
sourceTraceValidator	Untraceable values entering final document.
dateFieldLockValidator	Date of issue/birth/registration swapped across zones.
monthMapValidator	February/October and other month mapping errors.
numericEvidenceValidator	Unverified digits, dates, addresses, record numbers.
passportPerforationValidator	Wrong series/number from perforated IDs.
digitShapeComparator	8/0/1/6/9 visual confusion.
duplicateLanguageCollapser	Ukrainian/Russian duplicate output.
agencyNameResolver	Invented or modernized agency names.
adminTermValidator	сmt translated as city, oblast as state, etc.
controllingSpellingResolver	Name spelling mismatch across packet.
scopeCompletenessValidator	Partial document presented as full document.
versionRedlineValidator	Silent corrections without change history.
serviceClaimsValidator	Unsafe website/PDF/email claims.
paymentGateValidator	Final PDF/DOCX download before payment.
certificationRecordValidator	Certified package without signer record.

27. QA Evidence Table and Acceptance Gate

Field	Source label	Raw value	Output	Status
Document Type	Title	Паспорт громадянина України	Passport of the Citizen of Ukraine	PASS
Series	Perforation	AA	AA	PASS
Number	Perforation	123456	123456	PASS
Date of Birth	Дата народження	01 січня 1980	01 January 1980	PASS
Date of Issue	Дата видачі	10 лютого 2000	10 February 2000	PASS
Place Type	Місце народження	сmt ...	... urban-type settlement	PASS

No QA evidence table → no final PDF. No source trace for a critical field → no final PDF.

28. Product and Quality Metrics

Metric	Target / signal
Numeric field accuracy	≥ 99.9% on eval set; zero known wrong digits in final outputs.
Critical field accuracy	≥ 99.5% across names, dates, identifiers, agency names.
Forbidden phrase rate	0 in final bureau-style output.
Unjustified guess rate	0; every low-confidence field flagged.
Field source-trace coverage	100% for critical fields.

Human correction rate	< 5% after stabilization; corrections become regression tests.
User completion rate	Track by device, language, age proxy; investigate drop-offs at upload/review/payment/signature.
Manual review rate	Expected in pilot; reduce only after accuracy is proven.

29. Evaluation Set and Regression Tests

- 20+ internal passport booklet samples with perforated identifiers.
- 20+ birth certificates with act record numbers and registry offices.
- 20+ marriage/divorce/death certificates.
- 10+ residence-registration/address samples.
- 10+ military documents only for manual-review pipeline.
- 10+ diplomas/transcripts with tables before releasing tabular renderer.
- Edge cases: glare, skew, seal overlap, Ukrainian/Russian duplicates, old/new agencies, handwritten numbers, multiple date zones.
- Every fixed production error becomes a regression test with expected output and failing validator.

30. Privacy, Security, and Retention

- Encrypt uploads and generated files at rest; use TLS in transit.
- Minimize raw personal data in logs. Store field traces separately from user-facing output.
- Apply retention policy to raw images and final files. Provide deletion path.
- Do not place ad pixels or behavioral tracking on document upload/review/certification pages.
- Audit access to sensitive documents and final packages.
- Watermark or blur unpaid final-layout previews, but allow field-level verification cards before payment.
- Never expose document images in analytics, training logs, or public support tickets.

31. Service Claims Policy

Category	Copy policy
Forbidden	USCIS accepted; guaranteed; approved translation; certified by AI; instant certified translation; 100% accepted; official USCIS translation.
Allowed	Prepared for USCIS-style submission; AI-assisted draft for human review; You review and sign if competent; USCIS may request a corrected or third-party translation.
Where enforced	Website, checkout, PDF templates, emails, SMS/Telegram, FAQ, ads, sample documents.

32. Pilot Acceptance Criteria

Area	PASS condition
Functional	User can upload a passport/birth certificate, review extracted fields, correct fields, pay, sign certification, and download final PDF.
Accuracy	10/10 pilot passport samples pass numeric/date/identifier validation with no known wrong critical fields.
UX	Mobile flow completable by non-technical user; large fonts; no screen has more than one primary task.

Legal copy	No forbidden claims in site, email, checkout, PDF, or sample.
Payment	Final clean PDF/DOCX unavailable before payment; field-level review available before payment.
Evidence	Every critical field has source trace and evidence crop.
Certification	No certified package without certification record and signer action.
Regression	Known historical errors are covered by tests: wrong date zone, 8/0 perforation confusion, round-seal garbage, city vs urban-type settlement.

33. Implementation Roadmap

Phase	Scope
Phase 0 - Design freeze	Finalize copy policy, data contracts, field maps, and PDF template.
Phase 1 - Narrow MVP	Passport booklet + birth certificate; source trace; evidence viewer; certification record; payment gate; final PDF.
Phase 2 - QA hardening	Regression test bank; translation memory with correction classifier; source-to-final audit.
Phase 3 - Additional documents	Marriage/divorce/death certificates; address parser; improved glossary.
Phase 4 - Complex docs	Military, diplomas/transcripts, medical records with manual-review gate and tabular renderer.

34. Master Agent System Prompt

You are a professional Ukrainian-to-English document translation agent for official U.S. submissions.

Priority order:

1. factual accuracy
2. numeric accuracy
3. name consistency across the packet
4. official terminology
5. clean bureau-style formatting

Translate legally meaningful content only. Do not describe layout, decoration, page numbers, covers, image placeholders, seals by shape, signatures, or bilingual structure unless the source contains readable legal text that must be translated.

For every critical field, create a source trace: field, source label, source zone, raw value, normalized value, confidence, review flag.

Never guess. Never copy from previous drafts. Never modernize historical agencies. Never invent numbers, dates, names, addresses, document identifiers, or agency names.

For old Ukrainian passport booklets: use Ukrainian text as primary, collapse Russian duplicates, parse perforated series/number separately, compare ambiguous digits, preserve administrative terms such as urban-type settlement, and do not translate old MVS/militia structures as police.

Before final output, run all QA validators. If any critical field fails, return REVIEW REQUIRED instead of final document.

Final output must look like a professional translation bureau document, not an AI image description.

35. Risks and Controls

Risk	Level	Control
Wrong digit/series/passport number	Critical	Numeric evidence table, perforation parser, digit comparator, human review threshold.
Wrong date/month	Critical	Date-field lock, month map validator, back-check to raw source.
Name spelling mismatch	Critical	Controlling spelling resolver and packet-level identity report.
AI learns bad correction	High	Correction Classifier; review before global Translation Memory update.
Old agency translated as modern police	High	Historical agency registry and no-modernization rule.
Layout garbage in final document	High	Forbidden phrase validator and bureau-style renderer.
Russian duplicate overrides Ukrainian source	High	Ukrainian-primary and duplicate-collapse rules.
Partial document presented as full document	High	Scope label and page completeness validator.
AI draft presented as certified	High	Certification policy and signer responsibility levels.
User abandons flow	High	Evidence Viewer, Smart Retake anti-loop, manual/help fallback.
Over-engineering blocks MVP	Medium	Phase roadmap; defer complex docs until core passport/civil-doc flow works.

36. File Targets for Productization

```

/docs/translation/DOCUMENT_TRANSLATION_ENGINE_V5.md
/docs/translation/USCIS_TRANSLATION_STANDARD.md
/docs/translation/UKRAINE_PASSPORT_BOOKLET_RULES.md
/docs/translation/UKRAINE_GLOSSARY.yaml
/docs/translation/TRANSLITERATION_PRIORITY.md
/docs/translation/NUMERIC_ACCURACY_PROTOCOL.md
/docs/translation/TRANSLATION_QA_AND_VERSIONING_PROTOCOL.md
/docs/translation/SERVICE_CLAIMS_POLICY.md
/prompts/translation-agent-system.md

/apps/web/lib/translation/documentClassifier.ts
/apps/web/lib/translation/zoneExtractor.ts
/apps/web/lib/translation/numericAccuracy/dateFieldLockValidator.ts
/apps/web/lib/translation/numericAccuracy/monthMapValidator.ts
/apps/web/lib/translation/numericAccuracy/passportPerforationValidator.ts
/apps/web/lib/translation/numericAccuracy/digitShapeComparator.ts
/apps/web/lib/translation/sourceTraceValidator.ts
/apps/web/lib/translation/bureauStyleRenderer.ts
/apps/web/lib/translation/translationQaValidator.ts
/apps/web/lib/translation/paymentGateValidator.ts
/apps/web/lib/translation/certificationRecordValidator.ts
/apps/web/lib/translation/sourceToFinalAudit.ts
/apps/web/lib/translation/correctionClassifier.ts

```